drone in the swarm is equipped with onboard processors, navigation sensors, and communication modules that allow it to make local decisions and share status information with other drones in real time. This distributed autonomy ensures that the swarm can reconfigure itself dynamically if some units are lost or disrupted, maintaining mission continuity under contested conditions.

One of the notable characteristics of SWARM-100 is its focus on high-altitude and rugged terrain operations, reflecting India's emphasis on mountain warfare and border security along geographically challenging zones. The system is designed to launch rapidly, navigate variable wind and temperature conditions, and execute coordinated target engagements without requiring continuous manual control. The swarm can be deployed from ground vehicles, forward posts, or portable launch units, offering mobility and tactical agility.

The SWARM-100 program demonstrates India's efforts to strengthen operational capacity through indigenous innovation, reduce dependence on imported UAV systems, and develop autonomous capabilities suited to its unique strategic environment. At the same time, its deployment raises considerations related to rules of engagement, security of communication links, electronic warfare resilience, and responsible operational use in conflict scenarios. As swarm technology evolves, SWARM-100 is expected to serve as a foundation for larger, more capable, multi-purpose autonomous swarms integrated into India's joint warfare doctrine.

Figure 9.4 illustrates the Indian Army's SWARM-100 initiative, which deploys a coordinated autonomous swarm of one hundred drones for reconnaissance, precision engagement, and encirclement missions. A central command node or "mothership" UAV issues mission objectives, while each drone uses distributed intelligence and a shared communication mesh to